Evidence Appendix

Gator Quant Hacks 2026

Rui Jiang, Aniket Shukla, Marvin Gao, Andy Wu 4 October 2026

1 E1: Orbital Calculations

Propagation method. We use the supplied fixed two-body Kepler model with epoch $t = 0 =$ 2026-09-22 00:00:00 TDB. Positions are measured in AU in the Sun-centered, geometric, three-dimensional J2000 ecliptic frame. If t is elapsed time in days, our Python propagator advances the mean anomaly

$$M(t) = (M_0 + nt) \bmod 360^\circ.$$

After converting angles to radians, it solves Kepler's equation

$$E - e \sin E = M$$

by Newton–Raphson,

$$E_{k+1} = E_k - \frac{E_k - e \sin E_k - M}{1 - e \cos E_k},$$

starting from $E_0 = M$. Iteration stops when the Kepler residual is at most 10^{-15} (maximum 100 iterations). The orbital-plane coordinates are

$$x' = a(\cos E - e), \qquad y' = a\sqrt{1 - e^2} \sin E,$$

and are rotated by argument of perihelion ω, inclination i, and ascending node Ω to obtain (x, y, z). Hours are converted by $t_{\text{days}} = h/24$.

Epoch and later-time validation. At $t = 0$, all eleven bodies agree with the Section 3 check vectors. Against the six-decimal values printed in the brief, the largest component error is 4.68×10^{-7} AU (Mars), well inside the required 10^{-5} AU tolerance. Table 1 also gives positions at $h = 300$ ($t = 12.5$ days), demonstrating that positions are propagated rather than frozen.

	$\mathbf{r}(0)$ [AU]			$\mathbf{r}(300\,\text{h})$ [AU]		
Body	x	y	z	x	y	z
Mercury	-0.213426	-0.410281	-0.013955	0.056841	-0.454135	-0.042327
Venus	0.679067	-0.257950	-0.042725	0.725310	-0.012041	-0.042015
Earth	1.003581	-0.023693	-0.000003	0.982164	0.189352	-0.000023
Mars	0.247476	1.526869	0.025929	0.080249	1.560199	0.030728
Ceres	0.375786	2.653941	0.014791	0.244555	2.659223	0.039134
Jupiter	-3.438179	4.038309	0.060149	-3.510619	3.980878	0.062008
Saturn	9.271221	1.717485	-0.398962	9.254395	1.785838	-0.399487
Uranus	8.962475	17.253919	-0.052129	8.918382	17.274207	-0.051484
Neptune	29.839098	1.351785	-0.715470	29.836970	1.391181	-0.716238
Relay A	2.000000	2.000000	0.000000	1.907580	2.088334	0.000000
Relay B	-2.000000	2.000000	0.000000	-2.088334	1.907580	0.000000

Table 1: Propagation check at the epoch and at hour 300. Coordinates are in AU.

As a worked later-time example, Earth's supplied values give

$$M(12.5) = 256.14049475^\circ + (0.9863415430^\circ/\text{day})(12.5) = 268.46976404^\circ.$$

Solving for E, forming (x', y'), and applying the three rotations gives

$$\mathbf{r}_E(300\text{h}) = (0.982164,\ 0.189352,\ -0.000023)\ \text{AU}.$$

Moving-receiver light time. For every individual launch we solve for arrival time rather than using the receiver's emission-time position:

$$t_a - t_e = 8.317\ \|\mathbf{r}_R(t_a) - \mathbf{r}_S(t_e)\| \quad \text{minutes}.$$

We initialize using the receiver position at t_e and apply fixed-point iteration

$$t_a^{(k+1)} = t_e + 8.317\left\|\mathbf{r}_R(t_a^{(k)}) - \mathbf{r}_S(t_e)\right\|,$$

re-propagating the receiver each iteration. Iteration stops when $|t_a^{(k+1)} - t_a^{(k)}| < 1$ ms. Serialization and local-access delays are added separately in the communication traces; this calculation is the photon flight time.

A useful worst-direction example at the epoch is Neptune→Mercury. If Mercury were incorrectly frozen at its emission position, the distance would be 30.11230966 AU and the flight time would be 250.4440794 min. Solving with Mercury moving during flight gives $d = 30.10905834$ AU and

$$\tau_{N\to M} = 250.4170382\ \text{min},$$

a 1.6225 s earlier arrival. The fixed-point iteration converges in two iterations.

The return direction must be solved separately because both bodies have moved. For an illustrative immediate geometric reply emitted when the forward photon arrives, re-propagating Mercury as sender and Neptune as receiver gives

$$\tau_{M\to N} = 250.4170909\ \text{min},$$

with return arrival at $t = 500.8341291$ min. Freezing Neptune at the return-emission instant would instead give 250.4170646 min, a 0.0016 s difference. Thus neither the return time nor its path length is assumed equal to the forward value; every direction and launch time is recomputed independently.

2 Scenario Evidence: Traces, Reliability and Funding

2.1 S1: Conditional Scenario Traces

Shared setup. Five independent runs reset the same \$220,000 and 1,200 AH shares (opening accounts in E3); no assets are shared across runs or added. Hours are from the supplied epoch. Conditional traces remove random loss only: receiver motion, 0.10 AU solar exclusion, serialization, queues, sessions and maintenance remain. Flights overlapping B–Neptune $[2, 26)$ or B–Ceres $[240, 264)$ cannot launch. Spot sessions start cold at $h0$. Futures establish their initial Mars–Earth and Ceres–Earth A sessions at $h-4$, with no asset locks; setup counts in traffic and rolling quotas. Initial conditions and results match S2/S3. Probability and funding evidence below apply to these same runs.

Cash transfer (CT) and equity trade (EQ). CT moves Neptune saver's \$20,000 to its own Earth account, with no fee. EQ exchanges Mars seller's 100 AH for Earth investor's \$10,000, plus a \$10 buyer-paid clearing fee. At 1 s, gateways exclusively lock \$20,000 (CT), or \$10,010 and 100 AH (EQ). Remote orders use $O_s \to A \to O_{\text{Earth}}$ followed by the local clearing handoff. Spot finals originate at C_{Earth} through A, using a separate institution session; they do not reuse an O_{Earth} session. This charges routine finals to the design's 280-packet clearing allowance. Earth's local buyer order reaches clearing at 2 s; it waits for the remote seller lock.

Message/event: hours from reset	CT: Neptune–Earth	EQ: Earth–Mars
Client orders received; source locks applied	0.0003	0.0003
Remote SYN received at Earth	4.1749	0.5654
SYN-ACK received at remote origin	8.3499	1.1308
Final handshake ACK received at Earth	12.5246	1.6962
Remote ORDER_LOCK released at Earth gateway	12.5249	1.6965
Clearing receives lock: discharge/backed claims	12.5252	1.6968
Earth recipient's signed final applied	12.5254: $20,000	1.6970: 100 AH
Clearing-final SYN received remotely	16.6999	2.2622
SYN-ACK received at clearing	20.8744	2.8275
Final handshake ACK received remotely	25.0488	3.3929
Remote CLEAR_FINAL applied; **completion**	**25.0491**	**3.3932**: seller $10,000

Local clearing handoffs take 1 s. DATA waits for the final handshake ACK. Clearing acts only on verified reservations; full completion additionally requires spendable recipient value and matching source reconciliation (E3). Arrival alone is not financial acceptance.

One futures rule, fixed before the price scripts. Mars factory is long and Ceres miner short 10 belt-metal contracts, multiplier 100, entry $100, notional $100,000, maturity $h480$. C_{Ceres} releases a signed cent-denominated observation locally at $h0, h240, h480$. After 1 s, O_{Ceres} forwards it through A to O_{Earth}, then clearing receives it through a 1 s handoff. No other settlement receives an automatic price broadcast. Both parties initially lock $25,000; maintenance equity is $10,000. For *received* price P, cumulative mark and equities are

$$V = 1,000(P - 100), \qquad E_L = M_L + V, \qquad E_S = M_S - V,$$

where M_L, M_S are actual posted margins. At the interim mark, if losing equity is positive but below maintenance, a call restores it to $25,000 within 24 h of gateway receipt. Opening instructions preauthorize the cure from available cash; gateway processing takes 1 s. Nonpositive losing equity triggers immediate funded default at posted collateral. An unavailable cure cannot create credit or release existing backing. Calls/finals originate at C_{Earth} through B; top-up lock notices return through A. Interim marks earmark a claim *within* margin, without spendable payment. At maturity, settle the total entry-to-final mark once, limited to posted collateral; no additional interim cure is assumed. Gateways act on signed, versioned instructions, not future script values.

Rising (UP), falling (DN), and binding variation (DF). UP observes $100 \to 120 \to 125$; DN observes $100 \to 80 \to 75$. Each deviates 20% at the interim mark and 25% finally. DF changes initial margin to $15,000 per side, retaining the $120 interim observation and the same 1,000-unit exposure, as Design §6 specifies. The opening balance sheet is unchanged. Before received prices differ, both standard runs take identical actions. UP and DN are reported separately and receive equal weight.

Financial/message step: hours	**UP**	**DN**	**DF**
Both initial margins locked: UP/DN $50,000; DF $30,000	0.0003	0.0003	0.0003
Opening $100 observation processed	0.5570	0.5570	0.5570
Both order locks recorded; position opens	0.5657	0.5657	0.5657
Interim observation published at Ceres	240.0000	240.0000	240.0000
Interim observation processed at clearing	241.6241	241.6241	241.6241
Cumulative reference mark V	+$20,000	–$20,000	+$20,000
Losing equity before cure	$5,000	$5,000	–$5,000
Call received at losing gateway	266.0015: Ceres	244.0207: Mars	default
$20,000 cure exclusively locked	266.0018	244.0210	none
Cure recorded at clearing	266.5415	245.6978	none
Maturity / final observation published	480.0000	480.0000	terminated
Discharge; backed payout: UP/DN $25,000; DF $15,000	481.5842	481.5842	241.6241
Mars local payment/release applied	483.9097	483.9094	244.0210
Ceres local payment/release applied	484.0370	484.0373	266.0015
Both local finals applied; completion	**484.0370**	**484.0373**	**266.0015**

UP/DN remain open from $h0.5657$ to $h480$, then funded pending the received final price until $h481.5842$. Interim $20,000 plus final $5,000 equals $25,000 once. Losing margin is $45,000 after cure; remaining equity $20,000 prevents default. DF terminates after 241.0585 h open, paying $15,000 of collateral; the excluded $5,000 reference-mark excess is not debt. Later prices are irrelevant to DF. B–Ceres maintenance delays the UP call and DF's Ceres final; A carries the interim mark. All five runs end paid, including DF's capped default. E3 reports exposure, peak locks and asset-hours, including the $70,000/31.82% scale demonstration.

2.2 E2: Probability, Traffic and Capacity

Per-launch and four-launch bounds. Every emitted backbone packet, including replies, receipts and retries, has fresh loss probability $p(d) = 1-e^{-0.02d}$, using distance to the receiver's arrival position. For fixed Kepler ellipses, $d \le D = r_{s,\max} + r_{r,\max}$, where $r_{\max} = a(1+e)$. Both relays have radius $\sqrt{8} \approx 2.828427$ AU. Thus the following bounds cover every launch in either direction on A or B, including B–Neptune fallback. They bound random loss *after* a legal launch, not solar availability.

Settlement–A/B	D **(AU)**	$p_{\max}$	**All 4 forward lost**	**4 unconfirmed rounds***
Earth	3.8450	7.403%	$\le 0.00301\%$	$\le 0.04131\%$
Mars	4.4945	8.598%	$\le 0.00547\%$	$\le 0.07331\%$
Ceres	5.8150	10.980%	$\le 0.01454\%$	$\le 0.18550\%$
Neptune	33.1864	48.508%	$\le 5.5364\%$	$\le 29.1603\%$

Entries are rounded upward from unrounded orbital bounds. Four forward-copy losses satisfy $\Pr \le p_{\max}^4$; they prevent delivery. Sender abandonment is different: delivered bytes can remain unconfirmed if receipts fail. With timely receipt return, a round fails confirmation with $q_j = p_j + (1-p_j)p_{\text{receipt},j}$, giving the starred bound $[1-(1-p_{\max})^2]^4$. This bound assumes a receipt, if emitted and successful, can return before its hop timer. Queue or contact delay can invalidate that assumption; across such timing failures we use the conservative abandonment bound 1 and the event simulation below, rather than claiming p^4 for actual abandonment. Fresh random draws do not make shared maintenance or queue delays independent. An abandoned hop leaves locks/claims funded and delivery status unresolved; it does not reverse execution or authorize an unlock.

All communications and both quotas. Backbone application messages are CT order/final (2), EQ seller order/final (2), UP/DN two locks, three prices, call, cure notice and two finals (9), and DF two locks, two prices and two default finals (6). Local packets are client orders, oracle releases and authority handoffs; the preauthorized cure is processing, not an additional local packet. All routes have two links. For S sessions and D DATA messages, launches are $4(3S+2D)$: $2S$ each SYN/SYN-ACK/ACK, $2D$ DATA, $2D$ DATA-ACK, and $2(3S+2D)$ one-link hop receipts. Origins are $S+D$.

Run	Local	D/S	Origins	Backbone	All packets	Peak/24 h
CT	3	2/2	4	40	43	4
EQ	6	2/2	4	40	46	4
UP	11	9/7	16	156	167	6
DN	11	9/8	17	168	179	6
DF	8	6/5	11	108	116	5

Counts include pre-epoch setup and trailing controls. Idle expiry renews futures sessions, with an extra Mars cure session in DN. Every principal uses 0/12 direct packets; no direct-dependent deadline is claimed. Shared backbone use peaks at 6/600 origins/24 h. Automatic transport controls and retries remain in traffic and capacity totals. Emission gaps are at least 1 s; queues peak at 3 (CT/EQ) or 4 (UP/DN/DF), and DATA occupancy at 2/64. All runs complete one transaction, so all-packet efficiency equals the listed total.

Quota checks charge CT/EQ's two remote origins to that gateway and their two clearing origins to C_{Earth}. No routine spot origin is charged to O_{Earth}; every gateway stays within 35/24 h and clearing within 280/24 h. Local authority dispatch is included in the local-packet count.

A cure notice carries call-receipt and lock times together in its existing DATA packet (two FIR-64 records). Earth verifies the receipt-based deadline from received evidence. Financial IDs suppress replays; observation publication times order marks. Recorded terminal state rejects stale marks/cures. No extra packet or free remote knowledge is assumed.

Completion probabilities, averages and repeated failure. CT/EQ were recomputed with 2,000 independent trials each, seeds $1042026000, \ldots, 1042027999$. UP/DN retain 10,000 trials each, seeds through 1042035999. DF's revised amounts preserve its messages, default branch and transport path under each loss realization; its existing 10,000 transport trials therefore apply. Twenty paired old/new seed checks also give identical launch counts and completion times. Each actual launch draws its own loss, with recomputed receiver geometry; there is no additional whole-route trial. Maintenance, FIFO and timers remain: four hop launches with $R_h = 2t_{\text{hop}} + 1$ h, four endpoint attempts with $R_e = 2T_0 + 24$ h, then replacement sessions for unresolved operations. Packet lifetime is 30 days; financial obligations persist.

Run	Mean finish h	95th h	99th h	$\Pr(T \le t_0)$	Wilson 95%
CT	72.2349	115.9249	135.9469	0.45%	0.24–0.85%
EQ	4.4406	6.6431	8.2678	51.75%	49.56–53.93%
UP	485.9067	489.0492	490.6847	29.71%	28.82–30.61%
DN	485.9187	489.0487	490.8365	29.34%	28.46–30.24%
DF	266.7725	269.6604	269.9598	65.32%	64.38–66.25%

t_0 is each S1 conditional completion time; success counts are 9, 1,035, 2,971, 2,934, 6,532 (CT/EQ denominators 2,000; others 10,000). Mean sampling half-widths (95%) are 1.033, 0.057, 0.032, 0.033, 0.023 h, excluding model uncertainty. All trials finish before $h2000$. Sample maxima are 189.3272, 14.5236, 496.7362, 509.6669, 273.9168 h. The **true worst case is unbounded**: indispensable losses can miss any finite deadline. Receipt loss may conceal execution. Repeated failures prolong locks and asset-hours; recovery reconciles operation records without repeated financial effects. Missing prices leave maturity pending, and silence never unlocks. These samples do not establish saturated throughput or every failure history.

2.3 E3: Balances, Exposure and Capital

Opening assets, locations and liabilities. All amounts are NeoDollars; AH is the only share type. Assets start as spendable balances at the named gateway, owned by the named client, with no scenario liability, lien or institution-funded reserve.

Owner	Location	Cash	AH
Earth investor	O_{Earth}	100,000	0
Mars seller	O_{Mars}	0	1,000
Mars factory	O_{Mars}	50,000	0
Ceres miner	O_{Ceres}	50,000	0
Neptune saver	O_{Neptune}	20,000	200
All institutions	respective settlements	0	0
Total		**220,000**	**1,200**

An origin lock remains under its owner's gateway until clearing's verified transaction assigns the exact backing to a named beneficiary. Pending destination claims are recorded at Earth's authority and backed by the corresponding source lien; destination spendability requires its signed local final. After authoritative reassignment, an old origin record is an unavailable *mirror*, not another asset or another spendable pool. Clearing's settlement obligation equals the backed beneficiary claim. Fees become clearing's spendable property at Earth, debiting clients.

Value-move balances over time. CT has \$200,000 inactive cash and all 1,200 shares unchanged. The table tracks the saver's active \$20,000; at clearing its escrow becomes the saver's Earth-destination claim. CT's source mirror is retired only at completion. EQ has \$120,000 inactive cash and 200 inactive AH at Neptune. Entries for buyer/seller are available cash/AH; backing entries are unavailable cash/AH. Times are the corresponding S1 milestones.

Run / stage	Origin or buyer	Destination or seller	Backing	Fee
CT: opening	20,000	0	0	0
CT: lock / clearing claim	0	0	20,000	0
CT: Earth credit / source final	0	20,000	0	0
EQ: opening	100,000/0	0/1,000	0/0	0
EQ: both locks	89,990/0	0/900	10,010/100	0
EQ: atomic match / backed claims	89,990/0	0/900	10,000/100	10
EQ: buyer's local final	89,990/100	0/900	10,000/0	10
EQ: seller's local final	89,990/100	10,000/900	0/0	10

At EQ match, the seller's \$10,000 is backed by Earth cash escrow and the buyer's 100 AH by Mars share escrow. Claims replace escrow in the unique ledger. CT moves only cash; the saver's 200 AH stays at Neptune. No institution advances unfunded destination value.

Futures balances and obligations over time. In UP/DF, winner W is factory at Mars, loser L miner at Ceres; DN reverses them. Inactive cash is \$120,000 and all shares are unchanged. A_W, A_L are available client cash; M_W, M_L are retained margins at their gateways; C_W is the winner's distinct backed payout claim. Clearing records the contract and its liability at Earth. The interim \$20,000 claim is an earmark *inside* M_L, excluded from C_W.

State at S1 milestone	A_W	A_L	M_W	M_L	C_W	Fee
Opening	50,000	50,000	0	0	0	0
UP/DN: both locks / position open	25,000	25,000	25,000	25,000	0	0
UP/DN: interim \$20,000 earmark	25,000	25,000	25,000	25,000	0	0
UP/DN: cure lock / confirmation / maturity	25,000	5,000	25,000	45,000	0	0
UP/DN: final discharge	25,000	5,000	25,000	20,000	25,000	0
UP: winner final first	74,990	5,000	0	20,000	0	10
DN: loser final first	25,000	25,000	25,000	0	25,000	0
DF: initial locks / position open	35,000	35,000	15,000	15,000	0	0
DF: capped default, no cure	35,000	35,000	15,000	0	15,000	0
DF: winner final / both finals complete	64,990	35,000	0	0	0	10
UP/DN: both finals complete	74,990	25,000	0	0	0	10

Each row satisfies $120,000 + A_W + A_L + M_W + M_L + C_W + F = 220,000$. At UP discharge, for example, the active \$100,000 is $25,000 + 5,000 + 25,000 + 20,000 + 25,000$; the payout replaces \$25,000 of losing margin, not a second claim against the same units. UP ends factory/miner \$74,990/\$25,000; DN \$25,000/\$74,990; DF ends factory/miner \$64,990/\$35,000. Clearing owns \$10. DF's delayed Ceres final retires its stale source mirror without changing canonical balances. All final margins, escrows and claims are zero; other accounts are unchanged.

Exposure, exclusive backing and capital measures. CT principal-loss exposure is zero at both endpoints: the saver retains its funded value. EQ buyer loss can reach \$10,010 including its fee if the acquired shares become worthless; seller nonpayment exposure is zero, since 100 AH exchanges only against locked cash. For either futures side, one interim cure and cent prices permit maximum gross loss \$49,990: V changes in \$10 steps, the largest still-positive-equity cure adds \$24,990 to \$25,000, and a final zero-equity default can consume it all. The coarse cash ceiling is \$50,000 per side; enforceable loss is always capped by posted margin. UP/DN actually lose \$25,000 on the adverse side; DF caps the tested short loss at \$15,000. With its lower initial margin and the same one-cure target, its possible posted-loss ceiling is \$39,990; both initial owners are exposed symmetrically before the price is received. Winners pay the \$10 fee from their payout. Clearing bears zero unfunded counterparty exposure.

Exclusive locks precede origination; authoritative reassignment replaces backing. Earmarks and source mirrors add no assets. Every state conserves \$220,000 and 1,200 AH. Asset-hours are $\int_0^T L_a(t)\,dt$ over uniquely unavailable units, including pending claims; a stale mirror adds no encumbrance after destination spendability.

Run	Peak \$	Peak AH	\$-hours	AH-hours	Cash use	Peak/value
CT	20,000	0	250,503.19	0	9.09%	1.0000
EQ	10,010	100	33,945.85	169.68	4.55%	1.0010
UP	70,000	0	28,556,172.74	0	31.82%	0.7000
DN	70,000	0	28,999,617.54	0	31.82%	0.7000
DF	30,000	0	7,320,621.26	0	13.64%	0.3000

Cash use divides peak cash locks by \$220,000; EQ share use is $100/1,200 = 8.33\%$. Settled values are CT \$20,000, EQ \$10,000, and each futures run \$100,000 notional, as prescribed for capital efficiency (peak/value). At $h300$, each standard run has \$70,000 simultaneously encumbered by its open contract, exceeding the \$55,000 scale threshold. UP dollar-hours equal $50,000(t_W - 1/3600) + 20,000(t_L - t_{\text{cure}})$; DN cures earlier and therefore accumulates more dollar-hours. EQ integrates \$10,010 until match and the seller's \$10,000 claim until seller credit, plus 100 AH until buyer credit. CT integrates \$20,000 until Earth credit; DF integrates \$30,000 until winner credit. Receiver arrivals, maintenance, serialization, counts and these lock integrals are checked against the event records. These tables give the funding proof without requiring attachments.

3 S2: Stress and Financial Recovery

Scenario and incident selection. Hours are from the frozen epoch. Each run resets the E3 opening balance sheet. Only the S1 UP futures contract is active: \$25,000 initial margin per side, a \$20,000 Ceres cure, and one total \$25,000 final payoff. Institutions have no opening assets.

We select **Earth gateway isolation (Node 3),** $[h482.750, h554.750)$. For 72 hours, every launch touching Earth fails, including receipts; launches emitted earlier can still arrive. Durable records and local access survive. The incident and restoration are unannounced. Earth is the clearing hub: neither relay bypasses its isolation. The start catches settlement after the final backed claim is recorded but before local settlement completes.

Selection evaluated 5,556 conditional cases: all nine isolation gateways, all 11 forced-loss nodes and all 11 institution endpoints for resets, on a six-hour $h0$–$h480$ grid; then Earth/Mars/Ceres isolation at minute spacing in $h470$–$h485$ and within 1 ms of baseline emission boundaries. Earth and Ceres tie at $h560.9253$; Earth causes greater cash encumbrance, 33.925 versus 30.094 million dollar-hours. The coarse forced-loss maximum is $h509.0837$; coarse reset cases add no delay. Thus Earth is the *worst tested case*, not a continuous-time optimum; brief reset effects may be missed by the coarse grid. Incident selection was not

re-optimized under random loss.

Conditional trace and obligations. The trace below suppresses background random loss but retains incident loss, moving-receiver geometry, maintenance, serialization, queues and retries. Price and top-up records use Relay A; margin calls and settlement use Relay B. Local handoffs take 1 s. Session setup at $h-4$ is counted; without keepalives, seven-day expiry requires fresh handshakes at the later marks.

Hour	Received evidence, action and financial state
0.5657	Clearing has both funded orders; position opens with \$50,000 locked.
241.6241	Interim observation reaches clearing: Ceres' equity is \$5,000; a \$20,000 call is issued.
266.0015 / 266.5415	Ceres receives the call and locks the top-up 1 s later; clearing records it. Total margin is \$70,000.
481.5842	Final observation reaches clearing: future discharged; Mars owns a \$25,000 backed claim, not yet spendable.
482.750–554.750	Earth isolated. In-flight SYN-ACKs arrive at 483.1344/483.2192; Earth-touching ACK/data launches fail. No funds are released.
559.7591 / 560.1080	Fourth SYN-ACK attempts reach clearing after restoration; final ACKs and fourth data attempts follow.
560.5337 / 560.9253	Mars applies its credit and margin release; Ceres records its matching debit and releases residual margin. Both local updates are complete.

During isolation, the \$25,000 payment claim is backed by Ceres' collateral; Mars' \$25,000 margin and Ceres' \$20,000 residual margin remain locked. The interim top-up is already fulfilled, within the receipt-based 24-hour limit; no default occurs. The \$10 clearing fee is unpaid until Mars settles. These are all live obligations: no equity trade or transfer is active, and all shares remain unchanged. Local stale locks mirror the same backing; they are not additional assets and never expire merely through silence. Available cash \$150,000 plus unique locks/claims \$70,000 equals \$220,000; peak encumbrance is 31.818%. Final spendable cash is Earth \$100,000, Mars seller \$0, Mars factory \$74,990, Ceres miner \$25,000, Neptune \$20,000 and clearing \$10, with no remaining locks or claims.

Physical disconnection lasts 72 h. First financial service resumes at $h560.5337$; full recovery at $h560.9253$ is 6.1753 h after reconnection, 78.1753 h after onset, and 76.8883 h later than baseline. Both local financial updates are required.

Maintenance comparison and sensitivity. Removing both scheduled maintenance windows retains natural geometry, the incident, funding and guarantees. The falling-price sensitivity changes only observations to $100 \to 80 \to 75$; Mars then posts the top-up and pays Ceres under the same margin rule.

	Baseline	Isolation	No maint.	Falling
Call received (h)	266.0015	266.0015	244.1198	244.0207
Full recovery (h)	484.0370	560.9253	560.9253	560.9256
Cash encumbrance (10^6 dollar-h)	28.556	33.925	34.363	34.377
Origins / hop launches	16/156	16/226	16/226	17/238

All columns are conditional traces; "No maint." and "Falling" include the same Earth isolation. Removing B–Ceres maintenance delivers the call 21.8817 h earlier without changing recovery; it adds 437,634 dollar-hours because the same top-up locks earlier. B–Neptune has no scenario traffic. The falling run swaps final Mars/Ceres balances to \$25,000/\$74,990; the fee, total funding and peak locks are unchanged. Its unstressed completion is $h484.0373$, again giving 76.8883 h additional delay.

Additional forced-loss check (Design §10). A separate reset run puts a six-hour incident at Relay A during $[h240, h246)$, with maintenance retained and the same UP funding. Four emitted launches fail. The interim price reaches Earth at $h266.7015$, the call reaches Ceres at $h269.1929$, and completion remains $h484.0370$; the cured contract stays funded. It uses 16 origins, 160 backbone launches, and 28.492 million dollar-hours. This is a sensitivity run, not a second incident added to the isolation run.

Random loss: average recovery and delay probabilities. We additionally run 10,000 independent trials per configuration (40,000 total), from $h-4$, with $p(d) = 1-\exp(-0.02d)$ applied independently to *every*

non-forced launch, including data, hop receipts, handshakes, acknowledgments and retransmissions. Distance d is recomputed at each emission using the moving receiver. Incident launches fail with certainty; there is no additional whole-route loss trial. Trials retain all timers and constraints. Four launches are allowed per hop, and four endpoint attempts per operation; exhausted operations recover through replacement sessions. A cutoff at $h2000$ retains unfinished cases; all sampled trials finished.

Configuration	**Mean finish** h	**Mean** R	**95th** R	**99th** R
No incident	485.9067	—	—	—
Isolation	561.8439	7.09	9.67	11.05
Isolation, no maintenance	561.8392	7.09	9.67	11.10
Isolation, falling prices	561.8613	7.11	9.67	11.25

Here R is hours from reconnection at $h554.750$ to *full financial recovery*; percentiles are empirical, not bounds. The rising isolation mean has an approximate 95% Monte Carlo interval of 7.066–7.121 h. By the original conditional deadline, 4,746/10,000 recover (47.46%); within 12 h, 9,966 recover (99.66%; Wilson 95% interval 99.53–99.76%); within 24 h, 9,996 recover (99.96%). None exceed 48 h, giving a one-sided 95% upper bound of 0.02995% on that tail, rather than zero risk. These intervals quantify sampling error, not model uncertainty.

Worst case and recovery safety. The slowest rising trial finishes at $h588.0429$, **33.29 h after reconnection**, with 17 random failed launches and one replacement session. Mars settles at $h562.4731$; Ceres' \$20,000 residual margin remains locked until $h588.0429$. Replacement sessions occur in 13.69% of rising trials, counting recovery anywhere in the run. **The true worst-case recovery time is unbounded**: any finite deadline can be missed by sufficiently many indispensable launch failures with nonzero probability. Four-attempt limits bound a retry cycle, not financial obligation duration.

Recovery retains locks, reconciles durable status and rebuilds sessions without repeating financial execution. Missing final observations leave maturity pending; terminal state rejects obsolete marks and cures. All 40,000 sampled runs conserve \$220,000 and satisfy recorded limits. Conditional isolation has 56 forced failed launches, 14 abandoned hop copies, zero direct packets, at most six origins/24 h, queue four and window two. This tests one funded contract, not market saturation or all out-of-order histories; progress requires usable routes, capacity and persistent reconciliation.

4 S3: Access at Every Settlement

Authorities and access. Cash transfers, equities, equity options and covered shorts use the Earth master ledger. Futures and capped options use their contract-named price source; Ceres supplies the worked metal-price example. Its observations reach Earth through the gateway; clients need no separate feed. Pairs below give $h0/h300$. Letters A/B mean $s \to$ Relay A/B $\to$ the named gateway; L means a 1 s local handoff. Remote delays (minutes) include serialization/relay processing; client access and authority handoffs add 1 s each. L includes its handoff; handshakes are excluded. We enumerate all permitted simple routes of two or three links and choose the fastest launchable empty-queue route.

	To Earth		**To Ceres**		**Availability**
Settlement	**Via**	**min: 0 / 300**	**Via**	**min: 0 / 300**	**0 / 300**
Mercury	A	46.02 / 43.77	A	41.83 / 40.84	100 / 100%
Venus	A	40.57 / 37.67	A	36.37 / 34.73	100 / 100%
Earth	L	0.02 / 0.02	A	33.37 / 32.25	100 / 100%
Mars	A	33.91 / 33.44	A	29.71 / 30.50	100 / 100%
Ceres	A	33.37 / 32.24	L	0.02 / 0.02	100 / 100%
Jupiter	B	50.92 / 50.23	B	41.30 / 41.35	100 / 100%
Saturn	A	79.42 / 78.86	A	75.23 / 75.93	100 / 100%
Uranus	A	158.25 / 156.72	A	154.07 / 153.79	100 / 100%
Neptune	A	250.48 / 250.05	A	246.30 / 247.13	100 / 100%

Availability applies separately to each listed route: the fraction of $[h, h + 24)$ during which all its directed links can launch, including solar exclusion and maintenance, not random loss. Minute samples are

certified between samples by orbital-speed bounds: the minimum sampled photon clearance is 0.120887 AU and the maximum interpolation allowance is 1.19×10^{-5} AU, leaving clearance above 0.10 AU throughout. A is selected at Neptune; its B alternative has zero availability in $[0, 24)$ because photon flights overlap B–Neptune maintenance $[2, 26)$.

Comparable funded value moves. Earth-delay ranking is Earth, Ceres, Mars, Venus, Mercury, Jupiter, Saturn, Uranus, Neptune (unchanged at $h300$). Select ranks 1, 5 and 9. Each run resets the original \$220,000/1,200-share balance sheet, relocating only the \$100,000 buyer with its holdings to Earth, Mercury or Neptune. It buys 100 AH shares at \$100 from the same Mars seller. At 1 s the buyer locks \$10,010, including the \$10 clearing fee; the seller locks 100 shares. All remote orders and replies use A, as in S1. Orders start cold at the source gateway; replies originate at clearing and establish separate C_{Earth} sessions. The Mars SYN/SYN-ACK/final ACK reach their endpoints at $h0.5654/1.1308/1.6962$; its order reaches clearing at $h1.6968$ in every run.

Event (hours from reset)	**Earth: best**	**Mercury: median**	**Neptune: worst**
Buyer SYN reaches Earth	local	0.7673	4.1749
SYN-ACK reaches buyer	local	1.5347	8.3499
Final ACK / order at Earth gateway	local / 0.0006	2.3019 / 2.3021	12.5246 / 12.5249
Match, discharge and backed claims	1.6968	2.3024	12.5252
Buyer: 100 shares spendable	1.6970	4.6040	25.0491
Seller: \$10,000 spendable	3.3932	3.9991	14.2209
Full trade completion	**3.3932**	**4.6040**	**25.0491**
Origins / all backbone launches	4 / 40	8 / 80	8 / 80

At match, the buyer holds a backed 100-share claim and the seller a \$10,000 cash claim. Locks remain unavailable until finals. Every run ends buyer \$89,990/100 AH, seller \$10,000/900 AH, and clearing \$10; other holdings are unchanged. Funding conserves \$220,000 and 1,200 AH. Direct traffic/retries are zero; origins peak at eight/24 h, queue three and peer window one, within individual allowances. These are conditional times: Neptune–A launch loss is about 42.72% despite 100% launch availability. Failed finals leave funded claims pending for reconciliation.

5 E4: Long-horizon Orbital Access (Tier 3)

Scope and method. We scanned 200 Julian years, $t \in [0, 73050]$ days from 2026-09-22 00:00:00 TDB, using the frozen supplied Kepler elements and both circular relay orbits. The grid spacing was six hours: 292,201 epochs including both endpoints. Availability fractions use the 292,200 left-endpoint samples. All 38 directed backbone links were tested independently. The computed epoch positions match the brief's table within 4.68×10^{-7} AU per coordinate. No live ephemeris was used.

For each link we solved the moving-receiver equation

$$t_a - t_e = \frac{8.317}{1440} \, \|\mathbf{r}_r(t_a) - \mathbf{r}_s(t_e)\| \quad \text{(days; positions in AU)},$$

and tested the closest point on the emission-to-arrival segment against the 0.10 AU exclusion sphere. The iteration stopping criterion was 10^{-11} day, stricter than the required 1 ms. The two one-time maintenance windows were applied at each actual emission time; no additional S2 incident was imposed. For departures to Earth clearing and Ceres pricing, we evaluated $s \to A \to d$, $s \to B \to d$, $s \to A \to B \to d$, and $s \to B \to A \to d$. Each successive hop uses its own emission and arrival positions, including 1 s serialization per link and 1 s processing at each relay. The reported route is the fastest feasible immediate, empty-queue candidate; these are physical route delays, not handshake or financial-completion times.

All-settlement results. Link entries are sampled percentages, *settlement-to-relay / relay-to-settlement*; route ranges are sampled minutes.

Settlement	A out/in (%)	B out/in (%)	To Earth (min)	To Ceres (min)
Mercury	90.95/90.95	90.94/90.94	34.81–83.98	20.22–90.75
Venus	94.62/94.62	94.62/94.62	32.64–84.53	19.04–91.13
Earth	95.69/95.69	95.69/95.69	Local*	17.08–91.75
Mars	96.88/96.88	96.87/96.88	25.13–87.49	12.37–94.07
Ceres	99.49/99.49	99.69/99.70	17.08–91.76	Local*
Jupiter	98.34/98.34	98.36/98.37	33.01–110.15	22.97–105.57
Saturn	98.99/98.99	98.95/98.95	66.95–144.86	53.41–143.92
Uranus	98.73/98.73	98.73/98.74	144.66–227.41	134.22–229.53
Neptune	99.03/99.03	99.02/99.03	239.58–311.60	226.79–308.94

*Same-site entries are not backbone routes; local access remains the paper's explicit co-location assumption. Both directions of $A \leftrightarrow B$ were available at every sampled emission. For each of the 16 distinct-site source–authority pairs, at least one candidate route was feasible at every grid point. Thus sampled route availability, and the lowest four-sample estimate for a 24-hour departure window, were 100%. This does not establish continuous availability or survival of an already pinned session.

Sampling limits. An isolated continuous closure longer than six hours must contain a grid point; one lasting less than six hours can fit entirely between samples. There is no established positive minimum closure duration: a near-tangent photon segment can produce an arbitrarily brief crossing. Therefore we do not claim that six-hour sampling detects every closure, nor that the table contains exact extrema or exact time-integrated availability. A single transition's time is initially uncertain by up to six hours, and a detected interval's two boundaries by up to twelve hours in total. Repeated missed intervals can accumulate; this is not a global error bound. The boundary refinement below improves the named event only. These limitations apply to the Tier 3 scan claim.

Refined event and packets in flight. Mercury$\rightarrow A$ changes from clear to solar-blocked within the original bracket $[82.5, 82.75]$ days. Bisection narrows the entrance to $[82.619934260845, 82.619934268296]$ days, a 0.644 ms bracket. The exit, initially bracketed by $[93.5, 93.75]$, refines to $[93.731123812497, 93.731123819947]$ days; the detected closure spans approximately 266.669 hours. The day-$[83, 84)$ link window has zero available launches on an additional one-minute grid, illustrating that endpoint link service can be poor even when another route remains available.

A launch emitted one second before the refined entrance clears the Sun at 0.100000203 AU and arrives 27.165502 minutes later, *after* the entrance. Its original emission-to-arrival segment remains valid; the later link closure does not retroactively cancel it. A launch one second after the entrance has clearance 0.099999797 AU and is unavailable. Queued launches must therefore be tested at their own emission, and receipts independently at their later emission. Pinned-session recovery still follows transport rules rather than instantaneous rerouting.

Poorest route service found. The six-hour grid's longest best-route delay is Neptune$\rightarrow A \rightarrow B \rightarrow$Earth: 311.596052 minutes at day 27,092. A one-minute refinement over days $[27091.5, 27092.5]$ finds 311.598283 minutes at day 27092.011806 (about year 74.17), with a feasible route at every refined sample. This is the poorest delay found, not a certified global maximum. The corresponding 24-hour window is explicitly identified by those endpoints. On the coarse grid, both two-link candidates are unavailable at 288 Neptune-to-Earth departures (72 days of sample weight); three-link routing restores those samples. That weight is not a refined cumulative outage duration. The 264.27-minute comparison retained in Design §1 is not a validated full-route worst-case bound; the current scan finds 311.598283 minutes. The scan also does not validate the 85.77-minute hub mean or rank alternative clearing locations. Operationally, evaluate three-link candidates before session creation, retain locks while outcomes are unknown, and separately account for setup, queues, quotas, loss, and reconciliation in S1–S2.

Reproducibility and interpretation. Route selection is repeated for each hypothetical new departure; it does not authorize changing a live session's pinned path. Reverse link geometry is tested independently, but the table is not a round-trip completion model. Maintenance occurs only at its original hours, not periodically every year. Random loss is excluded from geometric availability; it remains a separate E2 probability and recovery issue.

6 E5: Shifted-Epoch Evaluation of the Published Design

Tier 3; unchanged financial baseline. At offsets 1, 10 and 100 Julian years and the difficult day 27,092, all bodies advance by the offset before each independent financial reset. Observations and maturity shift with scenario start; original maintenance and S2 incidents do not repeat. These are new conditional event runs of the published $25,000-per-side contract, including the $20,000 interim cure.

Spot orders start cold; their clearing-origin finals establish separate sessions. Futures initiate the two inbound order sessions at relative $h-4$, with no financial locks, exactly as S1. Seven-day expiry triggers counted new handshakes at later observations. Route selection compares both one-relay and both cross-relay simple routes at each new session; the chosen route stays pinned. This implements Design §8's checked session/replacement route choice within the brief's three-link limit. Relays and shared physical links remain unchanged.

Offset: years; days	Routes N / M / C	Delays (min)	Availability
1; 365.25	N-A-E / M-B-E / C-B-E	288.4 / 47.2 / 46.6	100%
10; 3652.5	N-A-E / M-B-E / C-B-E	259.6 / 41.9 / 40.5	100%
100; 36525	N-B-E / M-B-E / C-A-E	277.5 / 38.5 / 41.0	100%
74.174; 27092	N-A-B-E / M-B-E / C-B-E	311.6 / 59.0 / 67.8	100%

N/M/C/E denote Neptune/Mars/Ceres/Earth; A/B denote relays. Delays are initial one-way routes to clearing, not round trips. Availability is the fraction of five-minute samples over the following 24 hours for each listed route, not a continuous guarantee or packet-delivery probability.

Offset (yr)	Transfer	Equity	Position open	UP complete	DN complete
1	28.8430	4.7218	0.7869	484.8152	484.8149
10	25.9618	4.1973	0.6997	484.0846	484.0843
100	27.7462	3.8520	0.6845	484.0053	484.0053
74.174	30.5823	5.8982	1.1307	486.9913	486.9913

Times are relative hours; completion requires both local finals. At the difficult epoch, the outward transfer uses three links; its separately created return session can choose a later legal route. Key price-run events are:

Received event, relative hour	UP	DN
Position opens	1.1307	1.1307
Interim price received	243.4454	243.4454
Call received	246.8911	245.7955
Cure recorded	248.0404	248.1462
Final price / discharge	483.4955	483.4955
Mars final	485.8058	485.8058
Ceres final	486.9913	486.9913

At every offset, UP ends factory/miner $74,990/$25,000; DN reverses them; clearing earns $10. Margin peaks at $70,000 (31.82%); cures are timely and no run suspends. CT/EQ final balances match E3. All runs conserve $220,000 and 1,200 shares.

Origins are CT/EQ 4 each, UP/DN 16/17; rolling peaks stay within each gateway's 35 and clearing's 280. Direct traffic is zero, queue ≤ 3, window ≤ 2. Total backbone launches are CT/EQ 40 each (difficult-epoch CT: 50), UP/DN 156/168, including setup and trailing controls.

These results test Design §§3–9 and §12 at selected phases, not Earth's global optimality or uninterrupted service. Resets are independent experiments; missing prices/finals preserve named funded claims. Deployment quantities in Design Table 14 remain assumed extension inputs. Supporting references are NASA/JPL's DSN Handbook 810-005 (`deepspace.jpl.nasa.gov/dsndocs/810-005/`), NASA's Mars Fact Sheet (`nssdc.gsfc.nasa.gov/planetary/factsheet/marsfact.html`), and Shapiro, *Physical Review Letters* 13, 789–791 (1964), DOI 10.1103/PhysRevLett.13.789. These references do not establish the chosen speed, carrier frequencies or handoff assumptions as maxima.